

Empirical Risk Minimization I

Clément Lalanne

What this lecture does

Make the problem tractable. The 0-1 loss is piecewise constant. Convexification replaces it by a loss that can be minimized, at a cost that has to be stated.

Isolate one quantity. The excess risk splits into three pieces, and only one of them is a probabilistic problem, the uniform deviation $\sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)|$.

Bound it, twice. A union bound settles finite classes completely. The bounded differences method handles any class, but leaves an expectation behind.

That leftover expectation is the whole subject of the next lecture.

Setting and notation

Sample. $S = (z_1, \dots, z_n)$ with $z_i = (x_i, y_i)$, drawn i.i.d. from $\mathbb{P}$ on $\mathcal{X} \times \mathcal{Y}$, so $S \sim \mathbb{P}^{\otimes n}$. Subscripts on every expectation and supremum, abbreviated $\mathbb{E}_S$ for $\mathbb{E}_{S \sim \mathbb{P}^{\otimes n}}$ as in the concentration lecture.

Risks. $R(f) = \mathbb{E}_{(x,y) \sim \mathbb{P}}[\ell(y, f(x))]$ and $\widehat{R}_S(f) = \frac{1}{n} \sum_{i \leq n} \ell(y_i, f(x_i))$.

Predictors. A class $\mathcal{F}$, the empirical risk minimizer $\hat{f}_S \in \arg \min_{f \in \mathcal{F}} \widehat{R}_S(f)$, the best element of the class $f^\dagger \in \arg \min_{f \in \mathcal{F}} R(f)$, and the Bayes risk $R^\star = \inf_f R(f)$ over all measurable f .

The question. What can be said about $R(\hat{f}_S) - R^\star$, for a predictor chosen by minimizing the very quantity we are trying to control?

Part I. Convexification of the risk

The 0-1 loss is combinatorial

In binary classification, $\mathcal{Y} = \{-1, 1\}$, the main loss we care about is $\ell(y, \hat{y}) = \mathbf{1}\{y \neq \hat{y}\}$. The empirical risk is then piecewise constant in the parameter.

Two obstructions, already met in the introductory lecture: the gradient vanishes almost everywhere, so no descent method moves, and exact minimization over a linear class is NP-hard in general.

The remedy, in two steps. First reparametrise the classifier as the sign of a real valued function, which turns the risk into the expectation of a function of one real variable. Then replace that function, which is discontinuous, by a convex one.

Step 1: reparametrisation as a sign test

Write $f : \mathcal{X} \rightarrow \{-1, 1\}$ as $f(x) = \text{sgn}(g(x))$ for a real valued $g : \mathcal{X} \rightarrow \mathbb{R}$, with

$$\text{sgn}(u) = \begin{cases} 1 & u > 0 \\ -1 & u < 0 \\ \text{Unif}(\{-1, 1\}) & u = 0. \end{cases}$$

Remark. At $u = 0$ a function has been replaced by a randomised one, so the predictor now carries an auxiliary variable U , independent of everything else, and expectations are written $\mathbb{E}_{(x,y) \sim \mathbb{P}, U}$. In practice $\mathbb{P}_x(g(x) = 0) = 0$ and the case never occurs, but the convention is what makes the next computation an identity rather than an inequality.

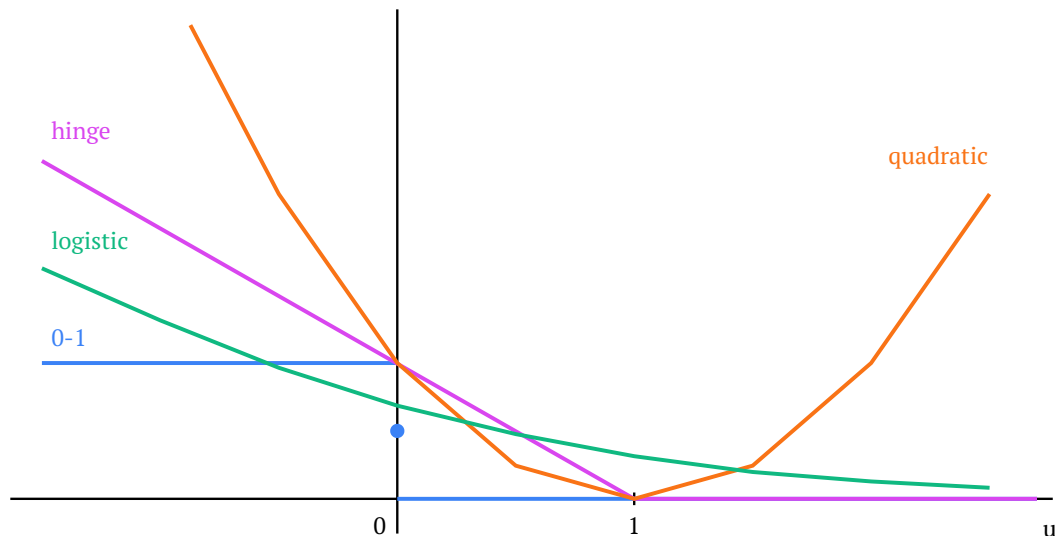
Step 1: the resulting identity

$$\begin{aligned} R(f) &= \mathbb{E}_{(x,y),U}[\mathbf{1}\{g(x) \neq 0\}\mathbf{1}\{f(x) \neq y\}] \\ &\quad + \mathbb{E}_{(x,y),U}[\mathbf{1}\{g(x) = 0\}\mathbf{1}\{f(x) \neq y\}] \\ &= \mathbb{P}_{(x,y)}(yg(x) < 0) + \frac{1}{2}\mathbb{P}_{(x,y)}(g(x) = 0) \quad \text{on } \{g \neq 0\}, f(x) \neq y \iff yg(x) < 0 \\ &= \mathbb{E}_{(x,y) \sim \mathbb{P}}[\Phi_{0-1}(yg(x))] \quad \Phi_{0-1}(u) = \mathbf{1}\{u < 0\} + \frac{1}{2}\mathbf{1}\{u = 0\} \end{aligned}$$

The risk of a classifier is the expectation of a fixed scalar function of the **margin** $yg(x)$. A positive margin means a correct prediction, and its size measures confidence. Everything that follows modifies Φ_{0-1} and nothing else.

Step 2: convex surrogates

Replace Φ_{0-1} by a convex $\Phi : \mathbb{R} \rightarrow [0, \infty)$ with better numerical properties, and minimize the **surrogate risk** $R_\Phi(g) = \mathbb{E}_{(x,y) \sim \mathbb{P}}[\Phi(yg(x))]$ instead.



Hinge $\Phi(u) = (1 - u)_+$, quadratic $\Phi(u) = (1 - u)^2$, logistic $\Phi(u) = \log(1 + e^{-u})$.

Step 2: what the surrogate costs

Domination. The hinge and quadratic losses satisfy $\Phi_{0-1} \leq \Phi$ pointwise, so $R(\text{sgn} \circ g) \leq R_\Phi(g)$ directly. The logistic loss does not: $\log(1 + e^{-u})$ equals $\log 2 < 1$ at $u = 0$. It dominates only after rescaling, and only on a compact.

The real question. Domination bounds the classification risk by the surrogate risk, but says nothing about the *excess risks*. Minimizing R_Φ is only legitimate if driving R_Φ to its infimum drives R to R^* , and that is a separate statement.

Why minimizing the surrogate is legitimate

Theorem (calibration). Let Φ be convex, differentiable at 0 with $\Phi'(0) < 0$. There is a nondecreasing convex $\psi : [0, 1] \rightarrow [0, \infty)$, continuous with $\psi(0) = 0$, such that for every g ,

$$\psi(R(\text{sgn} \circ g) - R^*) \leq R_\Phi(g) - \inf_{g'} R_\Phi(g').$$

Proof omitted, see Zhang (2004) and Bartlett, Jordan and McAuliffe (2006), where ψ is computed loss by loss.

The hinge loss gives $\psi(t) = t$, so the excess classification risk is at most the excess surrogate risk. The quadratic loss gives $\psi(t) = t^2$ and the logistic loss $\psi(t) \geq t^2/2$, so an excess surrogate risk ϵ transfers as $\sqrt{\epsilon}$ and as $\sqrt{2\epsilon}$: a slower transfer, and the price of a smooth loss.

Consequence. Convexification is not a change of problem. It is a change of objective for which the loss in the original objective is controlled.

From now on

After convexification the object being minimized is an empirical risk again, for a class of real valued functions g and the loss $\tilde{\ell}(y, g(x)) = \Phi(yg(x))$.

Parts II to V never use the structure of ℓ , only that it is bounded. They therefore apply verbatim to $\tilde{\ell}$, and we drop the tilde and write ℓ throughout. Part I is about making the minimization possible, everything that follows is about what the minimizer is worth.

Part II. Decomposition of the risk

Estimation and approximation

For the empirical risk minimizer $\hat{f}_S$ over a class $\mathcal{F}$,

$$R(\hat{f}_S) - R^* = \underbrace{\{R(\hat{f}_S) - \inf_{f \in \mathcal{F}} R(f)\}}_{\text{estimation error}} + \underbrace{\{\inf_{f \in \mathcal{F}} R(f) - R^*\}}_{\text{approximation error}}$$

The approximation error is deterministic, decreasing in the richness of $\mathcal{F}$, and belongs to approximation theory: it is not a statistical quantity and this course does not bound it. The estimation error is random, increasing in the richness of $\mathcal{F}$, and is what concentration is for.

Controlling the estimation error

$$\begin{aligned} R(\hat{f}_S) - R(f^\dagger) &= (R(\hat{f}_S) - \widehat{R}_S(\hat{f}_S)) + (\widehat{R}_S(\hat{f}_S) - \widehat{R}_S(f^\dagger)) + (\widehat{R}_S(f^\dagger) - R(f^\dagger)) \\ &\leq \underbrace{2 \sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)|}_{\text{uniform deviation}} + \underbrace{\sup_{f \in \mathcal{F}} (\widehat{R}_S(\hat{f}_S) - \widehat{R}_S(f))}_{\text{optimisation error}} \end{aligned}$$

Reading it. The first and third terms are each at most the uniform deviation. The middle term is the optimisation error, zero when the minimization is solved exactly, and in general as small as the training algorithm is willing to make it. It depends on the algorithm, not on the sample size, so we set it aside.

Everything statistical in empirical risk minimization is now inside one quantity, $\sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)|$. The rest of the lecture bounds it.

Part III. Concentration: bounded differences

Recall: the bounded differences method

Definition. $f : \mathcal{X}^n \rightarrow \mathbb{R}$ has bounded differences with constants $c_1, \dots, c_n$ if changing the i -th coordinate, the others being fixed and all points lying in the support, moves f by at most c_i .

Theorem (McDiarmid). For $X_1, \dots, X_n$ independent and f with bounded differences (c_i) ,

$$\mathbb{P}_S(f(S) - \mathbb{E}_S[f(S)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i \leq n} c_i^2}\right),$$

$$\mathbb{P}_S(|f(S) - \mathbb{E}_S[f(S)]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i \leq n} c_i^2}\right).$$

Proof. See the lecture on concentration inequalities for the full proof. Three steps: write $f(S) - \mathbb{E}_S[f(S)]$ as the sum of the increments of the Doob martingale $D_i = \mathbb{E}_S[f(S) \mid X_{1:i}]$; check that each increment lies, conditionally on the past, in an interval of length at most c_i and finally apply Hoeffding's lemma conditionally and deal with the exponential factors one at a time.

Recall: the two forms we will use

Deviation form. For every $\delta \in (0, 1)$, each of the following holds with probability at least $1 - \delta$:

$$f(S) - \mathbb{E}_S[f(S)] \leq \sqrt{\frac{1}{2} \sum_{i \leq n} c_i^2} \sqrt{\log(1/\delta)}, \quad |f(S) - \mathbb{E}_S[f(S)]| \leq \sqrt{\frac{1}{2} \sum_{i \leq n} c_i^2} \sqrt{\log(2/\delta)}.$$

Hoeffding, as the special case of a sum. If $Z_1, \dots, Z_n$ are i.i.d. with values in $[0, \ell_\infty]$ then $\frac{1}{n} \sum_i Z_i$ has bounded differences with $c_i = \ell_\infty/n$, so

$$\mathbb{P}_S \left(\left| \frac{1}{n} \sum_{i \leq n} Z_i - \mathbb{E}_Z[Z] \right| \geq t \right) \leq 2 \exp \left(-\frac{2nt^2}{\ell_\infty^2} \right).$$

Note the two logarithms. One sided costs $\log(1/\delta)$, two sided costs $\log(2/\delta)$. The factor 2 in front of the exponential is not decoration and will be kept everywhere below.

Part IV. A finite hypothesis class

Assumptions and the union bound

Hypothesis 1. $\ell(y, f(x)) \in [0, \ell_\infty]$ for every $f \in \mathcal{F}$ and every (x, y) in the support of $\mathbb{P}$.

Hypothesis 2. $|\mathcal{F}| < +\infty$.

$$\begin{aligned}\mathbb{P}_S(\sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)| \geq t) &\leq \sum_{f \in \mathcal{F}} \mathbb{P}_S(|\widehat{R}_S(f) - R(f)| \geq t) && \text{union bound, } \mathcal{F} \text{ finite} \\ &\leq \sum_{f \in \mathcal{F}} 2 \exp\left(-\frac{2nt^2}{\ell_\infty^2}\right) && \text{Hoeffding, } f \text{ fixed, Hypothesis 1} \\ &= 2|\mathcal{F}| \exp\left(-\frac{2nt^2}{\ell_\infty^2}\right) =: \delta\end{aligned}$$

Each $\widehat{R}_S(f)$ is an average of n i.i.d. variables in $[0, \ell_\infty]$, and f is fixed inside the sum, which is what makes the second line legitimate.

The uniform bound for a finite class

Theorem. Under Hypotheses 1 and 2, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathbb{P}^{\otimes n}$,

$$\sup_{f \in \mathcal{F}} |\hat{R}_S(f) - R(f)| \leq \frac{\ell_\infty}{\sqrt{2n}} \sqrt{\log \left(\frac{2|\mathcal{F}|}{\delta} \right)}.$$

Corollary. Combining with Part II, with probability at least $1 - \delta$,

$$R(\hat{f}_S) \leq \inf_{f \in \mathcal{F}} R(f) + \frac{2\ell_\infty}{\sqrt{2n}} \sqrt{\log \left(\frac{2|\mathcal{F}|}{\delta} \right)} + \epsilon_{\text{opt}}.$$

Reading the finite class bound

Rate. The deviation is of order $\sqrt{\log |\mathcal{F}|/n}$. Doubling the sample halves the squared deviation, whereas squaring the size of the class only doubles it.

Complexity appears logarithmically. This is the first quantitative form of the estimation and approximation tension. A richer $\mathcal{F}$ lowers $\inf_{f \in \mathcal{F}} R(f)$ and raises $\log |\mathcal{F}|$.

And this result is often useless. Every class used in practice is infinite: linear predictors, neural networks, a single real hyperparameter. The bound reads $+\infty$. The union bound is the step to blame, we show in the next section how to improve it.

Part V. From high probability to expectation

The supremum has bounded differences

Split the two sides, since the tools below are one sided:

$$\sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)| \leq \sup_{f \in \mathcal{F}} (\widehat{R}_S(f) - R(f)) + \sup_{f \in \mathcal{F}} (R(f) - \widehat{R}_S(f)).$$

Bounded differences. Replace one pair z_i by any other pair in the support. Under Hypothesis 1, each $\widehat{R}_S(f)$ moves by at most ℓ_∞/n , uniformly in f , and $R(f)$ does not move at all. A supremum of functions each moving by at most c moves by at most c , so both suprema above have bounded differences with $c_i = \ell_\infty/n$, hence $\sum_{i \leq n} c_i^2 = \ell_\infty^2/n$.

Proof. If $|u(f) - v(f)| \leq c$ for every f , then $u(f) \leq v(f) + c \leq \sup_{f' \in \mathcal{F}} v(f') + c$ for every f . The right side no longer depends on f , so it bounds $\sup_f u(f)$ as well: $\sup u \leq \sup v + c$. Exchange u and v . ■

The consequence

Proposition. Under Hypothesis 1, for every $\delta \in (0, 1)$, each of the following holds with probability at least $1 - \delta$:

$$\sup_{f \in \mathcal{F}} (\widehat{R}_S(f) - R(f)) \leq \mathbb{E}_S[\sup_{f \in \mathcal{F}} (\widehat{R}_S(f) - R(f))] + \frac{\ell_\infty}{\sqrt{2n}} \sqrt{\log(1/\delta)},$$

$$\sup_{f \in \mathcal{F}} (R(f) - \widehat{R}_S(f)) \leq \mathbb{E}_S[\sup_{f \in \mathcal{F}} (R(f) - \widehat{R}_S(f))] + \frac{\ell_\infty}{\sqrt{2n}} \sqrt{\log(1/\delta)}.$$

Proof. McDiarmid applied to each supremum, with the constants of the previous slide, in its one sided deviation form. ■

Where this leaves us

The uniform deviation is now concentrated around its expectation, at rate $n^{-1/2}$, **for any class whatsoever**, with no finiteness and no structure assumed.

But it is concentrated around an unknown number. Nothing above says that $\mathbb{E}_S[\sup_{f \in \mathcal{F}}(\widehat{R}_S(f) - R(f))]$ is small. For a rich enough class it is not: at $\mathcal{F}$ equal to all measurable functions and the 0-1 loss, that expectation equals 1 for every n .

Two routes. The union bound gives a complete answer and dies at $|\mathcal{F}| = \infty$. Bounded differences survives any class and leaves one expectation unevaluated.

Next lecture. Bound $\mathbb{E}_S[\sup_{f \in \mathcal{F}}(\widehat{R}_S(f) - R(f))]$ by the Rademacher complexity of the class, then compute that complexity for linear models.

What to remember

Convexification changes the objective, not the problem. The margin identity $R(f) = \mathbb{E}_{(x,y)}[\Phi_{0-1}(yg(x))]$ isolates one scalar function, and calibration bounds what replacing it costs.

Risk decomposition. Excess risk is approximation, plus optimisation, plus twice the uniform deviation $\sup_{f \in \mathcal{F}} |\widehat{R}_S(f) - R(f)|$.

Finite classes are finished. $\ell_\infty \sqrt{\log(2|\mathcal{F}|/\delta)/(2n)}$, by a union bound over the class and Hoeffding inside it.

Infinite classes are reduced, not solved. McDiarmid concentrates the uniform deviation around its mean for any class. The mean is the topic of the next lecture.